

Linear Combination and Spanning Sets

Definition 10.2.1. A vector $\mathbf{v}$ in a vector space V is called a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in V if $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$$

where $c_1, c_2, \dots, c_n$ are scalars.

Example 10.2.1. If $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\mathbf{v}$ is a linear combination of vectors in the set S where

$$S = \left\{ \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\mathbf{v}_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}}_{\mathbf{v}_2}, \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}_3} \right\}.$$

Solution. To see if $\mathbf{v}$ is a linear combination of vectors in S , consider the following equation:

$$\begin{aligned} c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 &= \mathbf{v} \\ c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ \begin{bmatrix} c_1 - c_3 \\ 2c_1 + c_2 \\ 3c_1 + 2c_2 + c_3 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

The augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array} \right] \quad [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \mid \mathbf{v}]$$

Row reduction gives

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array} \right] \xrightarrow{\substack{-R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 4 & -2 \end{array} \right] \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The reduced row echelon form shows that c_3 is free. Therefore, if $c_3 = 1$, then

$$c_1 = 2, \quad c_2 = -3.$$

Therefore, $\mathbf{v} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3$.

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Remark This linear combination is not unique. For example, if $c_3 = 0$, then $c_1 = 1, c_2 = -1$. This gives $\mathbf{v} = \mathbf{v}_1 - \mathbf{v}_2$.

Example 10.2.2. Show that $\mathbf{v} = \begin{bmatrix} 0 & 8 \\ 2 & 1 \end{bmatrix}$ is a linear combination of the vectors in the set

$$S = \left\{ \underbrace{\begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}}_{\mathbf{v}_1}, \underbrace{\begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix}}_{\mathbf{v}_2}, \underbrace{\begin{bmatrix} -2 & 0 \\ 1 & 3 \end{bmatrix}}_{\mathbf{v}_3} \right\}.$$

Solution. We want to solve the vector equation for c_1, c_2 and c_3 .

$$\begin{aligned} c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 &= \mathbf{v} \\ c_1 \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} + c_3 \begin{bmatrix} -2 & 0 \\ 1 & 3 \end{bmatrix} &= \begin{bmatrix} 0 & 8 \\ 2 & 1 \end{bmatrix} \\ \begin{bmatrix} -c_2 - 2c_3 & 2c_1 + 3c_2 \\ c_1 + c_2 + c_3 & 2c_2 + 3c_3 \end{bmatrix} &= \begin{bmatrix} 0 & 8 \\ 2 & 1 \end{bmatrix} \end{aligned}$$

The system becomes

$$-c_2 - 2c_3 = 0$$

$$2c_1 + 3c_2 = 8$$

$$c_1 + c_2 + c_3 = 2$$

$$2c_2 + 3c_3 = 1$$

Therefore, the reduced row echelon form may be obtained as follows:

$$\begin{aligned} \left[\begin{array}{ccc|c} 0 & -1 & -2 & 0 \\ 2 & 3 & 0 & 8 \\ 1 & 1 & 1 & 2 \\ 0 & 2 & 3 & 1 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 0 & 8 \\ 0 & -1 & -2 & 0 \\ 0 & 2 & 3 & 1 \end{array} \right] &\xrightarrow{-2R_1 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -2 & 4 \\ 0 & -1 & -2 & 0 \\ 0 & 2 & 3 & 1 \end{array} \right] \\ &\xrightarrow{\substack{R_2 + R_3 \rightarrow R_3 \\ -2R_2 + R_4 \rightarrow R_4}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & -4 & 4 \\ 0 & 0 & 7 & -7 \end{array} \right] &\xrightarrow{\substack{\frac{1}{4}R_3 \rightarrow R_3 \\ \frac{1}{4}R_4 \rightarrow R_4}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \right] &\xrightarrow{-R_3 + R_4 \rightarrow R_4} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

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$$\xrightarrow{-R_2+R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -2 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\begin{array}{l} 2R_3+R_2 \rightarrow R_2 \\ -3R_3+R_1 \rightarrow R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The solution is given by $c_1 = 1$, $c_2 = 2$, $c_3 = -1$.

$$\therefore \mathbf{v} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3.$$

Non-Example. The vector $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$ is not a linear combination of vectors in the set S of

Example 10.2.1.

To see this—we consider the following equation:

$$\begin{aligned} c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 &= \mathbf{v} \\ c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \end{aligned} \tag{1}$$

As in the previous example, the augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array} \right] \quad [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \mid \mathbf{v}]$$

Row reduction gives

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array} \right] \xrightarrow{\begin{array}{l} -R_1+R_2 \rightarrow R_2 \\ -3R_1+R_3 \rightarrow R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 2 & 4 & -5 \end{array} \right] \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 3 \end{array} \right]$$

The last row shows that the eq. (1) has no solution. Therefore, there exist no scalars c_1, c_2 and c_3 such that the eq. (1) holds true.